High-level Production Test Plan

Subject: Vision Series

Author: Jonathan Vanden Berk (Magics Technologies)

version	Date	Reviewed by	Approved by
1.4	2026-04-28	...	...

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1 Revision History

Revision	Date	Description	Author
1.0	2026-04-10	Initial draft of the Vision Series Test Plan	Jonathan Vanden Berk
1.1	2026-04-13	Updated diagrams and test stage descriptions	Jonathan Vanden Berk
1.2	2026-04-15	Added product specifications and acceptance criteria	Jonathan Vanden Berk
1.3	2026-04-22	Tightened USLs for MAG-IMG002x1-NC bad-row/bad-column (T0035) and PRNU global outliers (T0038), and added PROD/EM/reject bin classification to the pixel-array defect and hot-pixel/DSNU/PRNU outlier parameter descriptions	Jonathan Vanden Berk
1.4	2026-04-28	Aligned MAG-IMG002x1-NC datasheet specifications with new datasheet rev 2.0 (April 2026): updated electro-optical typical values (conversion gain, SNR_MAX, DSNU, PRNU, DC, DCNU, read noise, saturation limit), added system gain (K) and column FPN (CFPN) rows, switched current-consumption test condition reference from FHD10 to FHD25, added the new datasheet pixel-array defect specifications table (section 7.5), removed the internal T0035/T0038 detailed parameter tables, refreshed all datasheet references in chapter 6, and extended the References chapter with traceability tables for the per-product test-plan documents and the GIT repositories holding each product's test-system software and Product specification and test Limits-file (pinned by repo, branch and last-touching commit on the Limits-file)	Jonathan Vanden Berk

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2 Introduction & Scope

This document outlines the verification framework for Magics Technologies' Vision Series products to ensure all quality targets are met. The Vision Series encompasses a diverse range of hardware, including standalone radiation-hardened ICs (Image Sensor (MAG-IMG002x1-NC), Serializer (MAG-CXP00002-NP), and DC/DC Converter (MAG-PSU00001-NP)) and fully integrated electronic Vision Modules (MAG-VIS100xx-N).

This test plan details the test flows, conditions, and stages, alongside the production and trimming procedures required to guarantee the reliability and performance of each product. Furthermore, this document defines the direct relationship between datasheet specifications and the specific test parameters used to verify them.

This document specifically covers the production screening and verification procedures performed on every manufactured unit to guarantee zero-hour quality and compliance with datasheet specifications.

Long-term reliability (e.g., High-Temperature Operating Life, thermal cycling, mechanical shock) and radiation tolerance (TID, SEE) are guaranteed by design and verified through separate periodic qualification campaigns. These are documented in dedicated product qualification reports and are outside the scope of this production test plan.

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3 Vision products verification process

The verification and testing of the Vision Series products follow a three-step horizontal process rail. This ensures that quality is maintained incrementally from the IC component level up to the fully integrated module. While testing a MAG-VIS100xx product spans from component to module, our verification is gated: customers purchasing IC-only or PCB products receive components that have successfully cleared the specific segments of the rail required for those tiers.

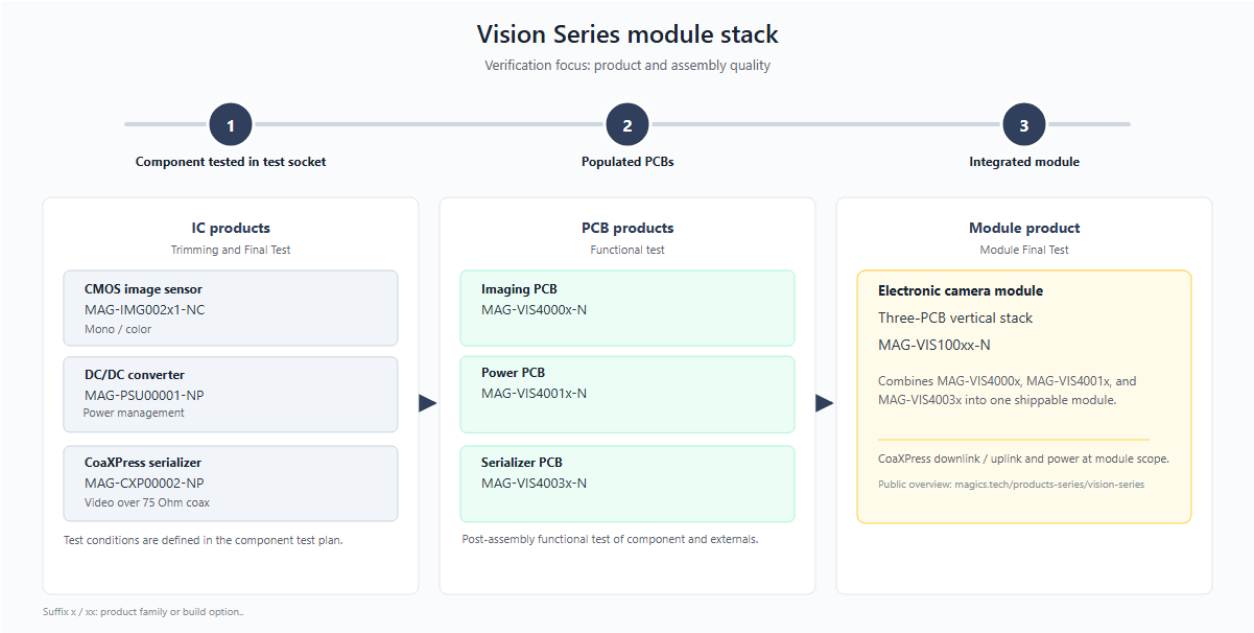

Figure 1: Vision Series: three-step verification rail

3.1 Level 1: Component verification

Focus: IC products are tested in test sockets across test conditions as specified in their individual test plans. These production tests focus on maximizing test coverage to ensure the early detection of any manufacturing defects.
Products: MAG-IMG002x1-NC, MAG-PSU00001-NP, MAG-CXP00002-NP.
Description: This level covers Trimming and Final Test (FT) of the individual integrated circuits. Electro-optical and image sensor array defect testing is included for the MAG-IMG002x1-NC. Testing procedures include continuity checks, I/O pin tests, interface tests, and parametric testing (including optional trimming). Functional verification and application-specific testing are also performed in addition to tests specific for the product family.

3.2 Level 2: PCB verification

Focus: Functional verification of assembled PCB products
Products: MAG-VIS4000x-N (Imager PCB), MAG-VIS4001x-N (Power PCB), MAG-VIS4003x-N (Serializer PCB).
Description: This level covers the functional testing of post-assembly boards prior to stack integration. Verifies that the soldering and board-level interconnects are reliable and that the individual sub-systems function correctly.

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3.3 Level 3: Module verification

Focus: Verification of the fully integrated module
Products: MAG-VIS100xx-N (Three-PCB vertical stack).
Description: This level covers the Module Final Test (MFT). It verifies the CoaXPress downlink/uplink and power delivery across the entire module scope. This final product level test validates high-speed timing and synchronization between the different components, ensuring any timing errors are caught at full operational speed.

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4 Test specifications

While the previous chapter outlined the high-level verification of the Vision Series products along the horizontal process rail, this chapter provides the detailed test specifications for each product, organized by product level test.

Note that the test procedures listed in this chapter measure significantly more parameters than those published in the product datasheets. The additional data collection serves Magics internal process monitoring: it enables detailed analysis of block and function-level performance on each IC, tracking of parametric drift over production lots, and early detection of process shifts. All internal parameters have defined limits and are actively monitored during production. To keep this report concise, only the parameters that are specified in the product datasheets are discussed in the acceptance criteria of Chapter 5.

4.1 Level 1: IC Test Specifications

This section covers the production and trimming tests for the individual integrated circuits. ICs are tested in test sockets across multiple conditions, focusing on maximizing test coverage for early detection of manufacturing defects.

4.1.1 MAG-IMG002x1-NC (Image Sensor IC)

4.1.1.1 Test flow

The production and trimming test flows for the image sensor are depicted below. The production flow covers electrical verification, visual inspection, and optical testing. A separate trim flow is used to calibrate internal references and timing settings. This trim flow is performed on a subset of each wafer batch, from this subset the best fit values for the batch are calculated and the DCF files are generated.

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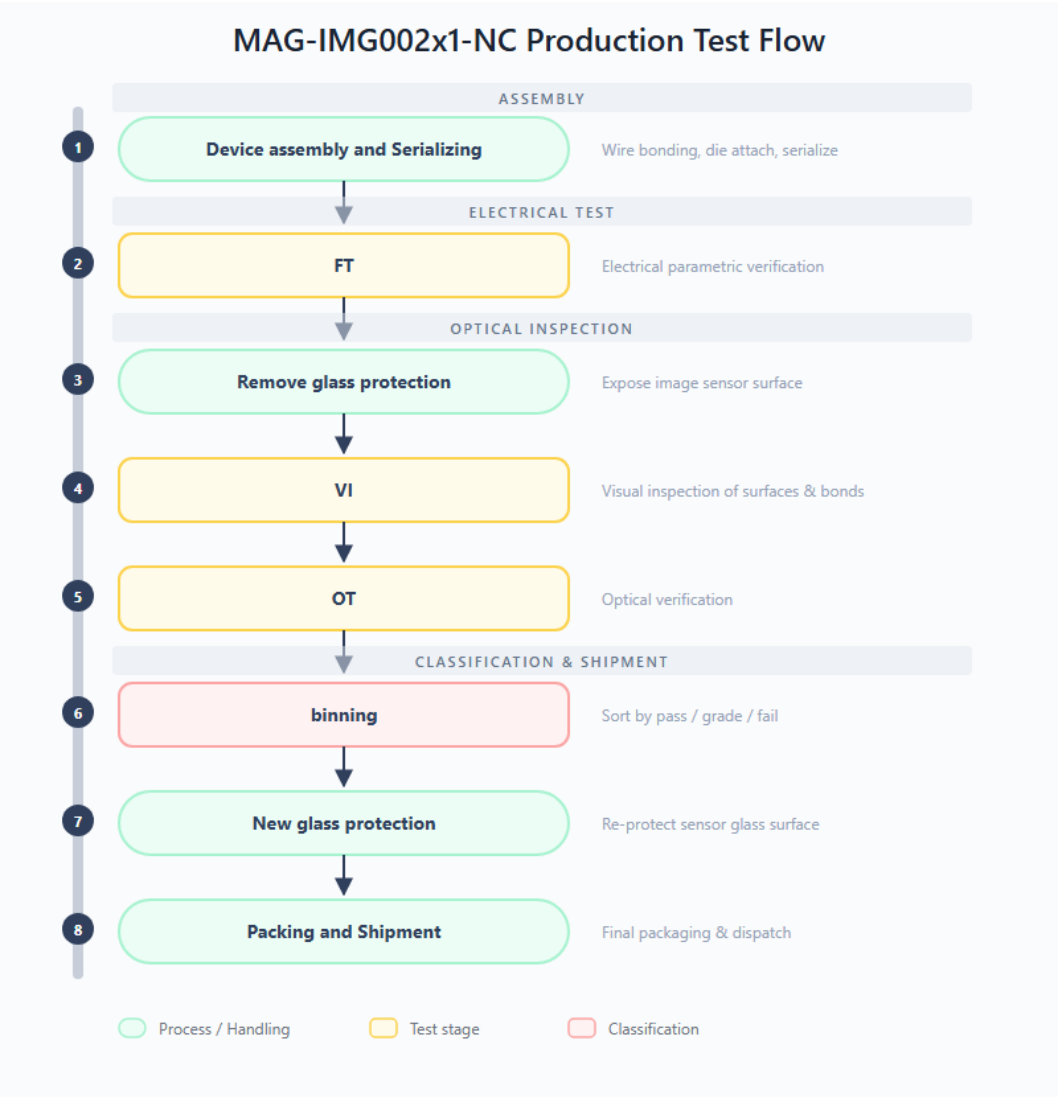

Figure 2: MAG-IMG002x1-NC Production Test Flow

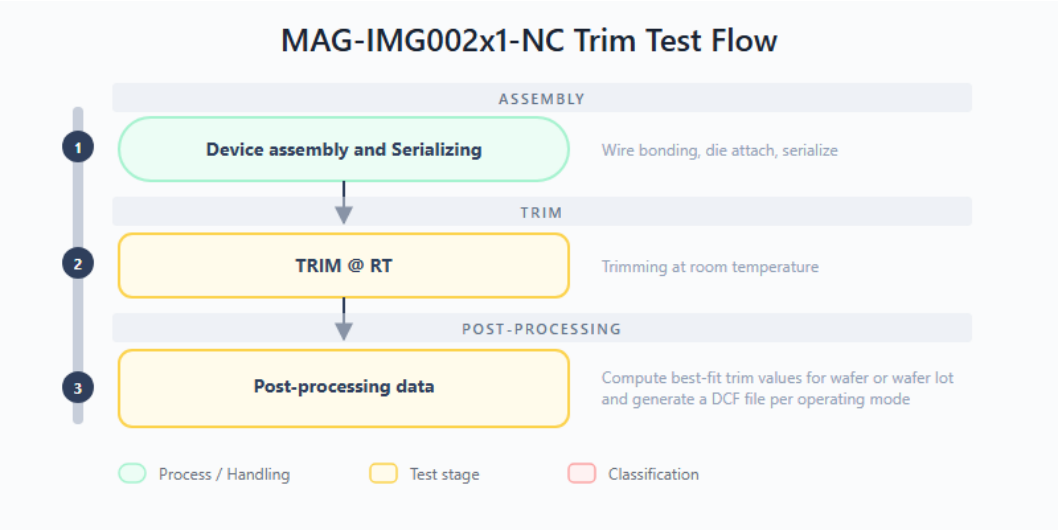

Figure 3: MAG-IMG002x1-NC Trim Test Flow

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4.1.1.2 Test conditions

The tables below define the temperature and supply voltage test conditions used across the IMG002x1 electrical and electro-optical test stages. Each test stage references a subset of these conditions (Min, Nom, and/or Max). The different supply domains are coupled, meaning that, for example, if the Max supply condition is chosen for a certain test stage, all supply domains will be set to their Max values. Cross-corners settings such as Analog at Vnom and Digital at Vmax are not allowed.

Table 1: MAG-IMG002x1-NC Temperature Conditions

Condition	Symbol	Min	Nom	Max	Unit
Temperature	Temp	0 (CT)	25 (RT)	85 (HT)	°C

Table 2: MAG-IMG002x1-NC Supply Voltage Conditions

Condition	Symbol	Min	Nom	Max	Unit
Digital supply voltage	VDDD	1.62	1.8	1.98	V
Analog supply voltage	VDDA	3.135	3.3	3.63	V
IO supply voltage	VDDIO	3.135	3.3	3.63	V
Pixel reset voltage	VD_RST	3.135	3.3	3.63	V

In the test stage section below, the default test conditions for each stage are listed. Sometimes there is a need to include or exclude specific conditions for a particular test procedure, this will be noted in the test procedure section. If the Conditions field is empty, the default test conditions apply.

4.1.1.3 Test stages

Production test stages:

FT (Final Test): Confirms that the product is compliant against test limits. This stage performs full electrical parametric verification of the image sensor, including continuity, power supply integrity, I/O thresholds, output drive strength, and power consumption measurements. Test conditions are: CT/RT/HT and Vmin/Vnom/Vmax, FHD10 mode, dark current related parameters are calculated from a ROI of 400 x 400 pixels in the center of the array. The sensor array is covered from any light input.
VI (Visual Inspection): Examination of external surfaces, glass lid, and wire bonding. This inspection identifies visible defects such as cracks, delamination, contamination, or bond wire anomalies before the device proceeds to optical testing.
OT (Optical Test): Verification of the optical performance of the image sensor. This stage validates dynamic range, pixel response uniformity, dark current, and other electro-optical parameters. The pixel array is also tested for defects in pixels, rows, and columns. Test conditions are: RT (ambient)

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and Vnom, FHD10 mode, dark current related parameters are calculated from a ROI of 400 x 400 pixels in the center of the array.

Trim test stages:

TRIM @ RT: Trimming at room temperature. Internal references are calibrated to their target values by applying and verifying trim codes at room temperature. Test conditions are: RT and Vnom.
Compute best-fit trim values: Post-processing of measured trim data to compute best-fit trim values per wafer or wafer lot and generate a DCF file per mode.

4.1.1.4 Test procedures

The table below lists all test procedures applicable to the MAG-IMG002x1-NC and indicates in which test stage each procedure is executed. The FT stage provides full electrical parametric coverage, while TRIM focuses on calibrating internal references. The OT stage validates optical performance under controlled illumination. The Conditions column notes any deviation from the default test conditions for that test stage, an empty cell means the default conditions listed under Test stages section applies.

Table 3: MAG-IMG002x1-NC Test Procedures

Test procedure	Description	Conditions	FT	TRIM	OT
Continuity	Each pin is connected to an SMU with power supplies off. A small current is forced into the pin, and the resulting voltage drop over the ESD structure is measured to verify proper pin connectivity.		X	X	X
Power Short	Checks for shorts on the power rails by supplying a very low voltage and measuring the resulting current per power domain.		X	X	X
Leakage	Measures the leakage current of each input pin or tristate output pin.		X	X	
VIL/VIH	Checks the input hysteresis of all digital input pins by sweeping the input voltage and observing the digital state transition.		X		
VOL/VOH	Measures the low and high voltage of digital outputs at 1 mA load.		X		
IOL/IOH	Measures the low and high level output current of digital outputs.		X		
IDDD	Measures the current consumption of the DUT in multiple states, video modes, frame rates and test patterns.	FHD10/HD40/ VGA50/FHD25	X	X	
Test buffer offset	Measures the offset of both test buffers on the test mux.		X	X	

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Static signal test	Measures the voltage/current of certain internal signals.		X		
PLL open-loop	Measures the open-loop frequency of the VCO while forcing the VCO voltage.		X		
PLL lock	Checks the lock status and measures the PLL-lock frequency.		X		
Ramp gen test	Measures the delay of the readout circuitry.		X		
POR	Validates the power-on reset behaviour of the device.		X		
Trigger mode	Generates an external trigger pulse and validates the DUT responds correctly.		X		
Testpattern generator	Tests the internal testpattern generator by comparing a captured image with a predefined reference.		X		
Serial interface	Validates the SPI serial interface functionality.		X	X	X
Trim test	Trims each internal block (voltage reference, current reference) by sweeping trim codes and selecting the best-fit value.			X	
Ramp gen trim	Trims the ramp generator circuitry to target voltage and slope values.			X	
Frame modes	Validates the different video modes by capturing and verifying images.	FHD10/HD40/ VGA50/FHD25			X
Blackpixel readout	Tests the black reference pixel readout.				X
Optical init	Defines the exposure time range required for correct linear fitting in DC and PTC analysis, and measures light nonuniformity.				X
Optical defect map	Captures raw images to map bright/dead pixels, rows and columns.				X
Optical PTC capture	Captures raw images necessary for Photon Transfer Curve (PTC) and spatial nonuniformity (SN) analysis.				X
Optical DC capture	Captures raw images necessary for dark current (DC) per pixel analysis.				X
Optical test processing	Reads raw images from previous tests and performs PTC, SN and DC per pixel analysis. Results are combined and reported to the results database.				X

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The table below maps each test procedure to the datasheet parameters it verifies. Parameters are listed by their Chapter 5 name. Note that only a subset of the test procedures is used to cover all datasheet parameters.

Table 4: MAG-IMG002x1-NC Parameter Traceability

Test procedure	Test ID	Parameters measured
VIL/VIH	T0004	TRIG_IN input threshold low, TRIG_IN input threshold high, TRIG_IN input hysteresis, RSTB input threshold (no hysteresis), SPI input threshold low, SPI input threshold high, SPI input hysteresis
VOL/VOH	T0005	SPI MISO output voltage (no load), VOH (no resistive load), VOL (no resistive load)
IDDD	T0007	VDDD current (active, FHD25 10-bit), VDDA+VDDIO+VD_RST current (active, FHD25 10-bit)
Static signal test	T0010	Bandgap reference voltage (VDD12BG)
Frame modes	T0024	Pixel output clock speed, Data output line speed
Optical defect map	T0035	Number of dead pixels, Number of bright pixels, Number of bad columns, Number of bad rows, Distance between bad columns, Distance between bad rows
Optical test processing	T0038	Conversion gain, System gain (K), Read noise, Dynamic range, SNR_MAX, Dark current (DC), Dark current non uniformity (DCNU), Dark signal non uniformity (DSNU), Photo response non uniformity (PRNU), Column Fixed-Pattern Noise (CFPN), Saturation limit

4.1.2 MAG-CXP00002-NP (CXP Interface IC)

4.1.2.1 Test flow

The production test flow for the serializer IC is depicted below. After device assembly, the chip undergoes trimming and OTP programming, followed by a comprehensive final test (FT). Trimming is done per part.

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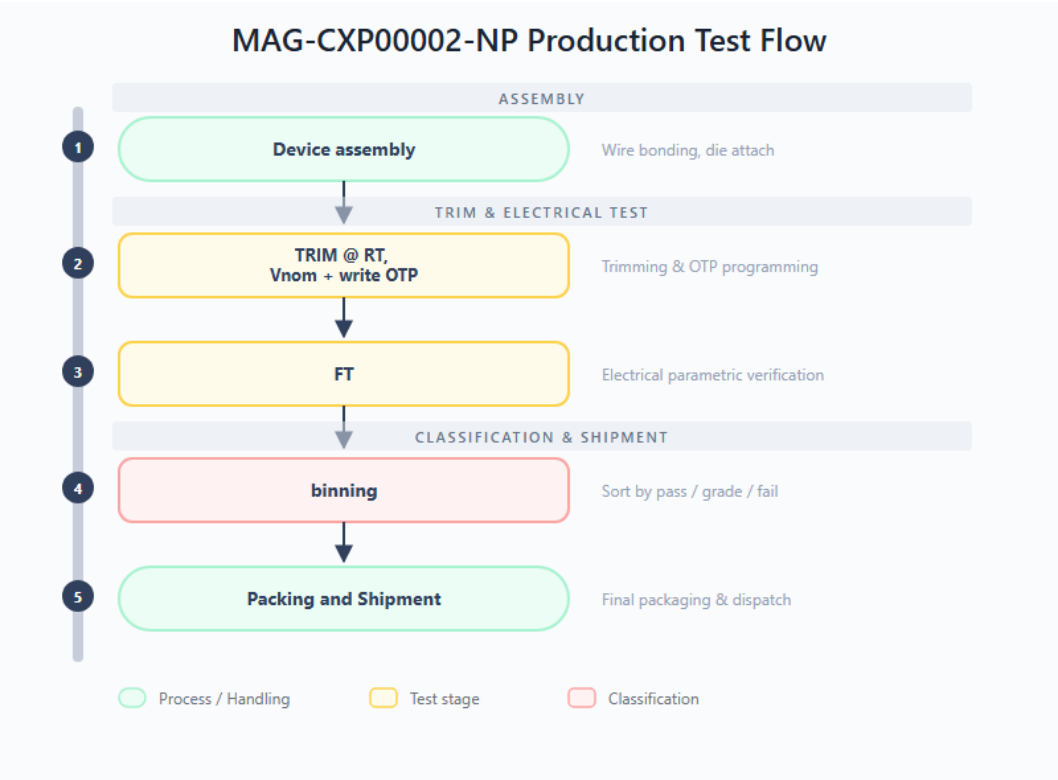

Figure 4: MAG-CXP00002-NP Production Test Flow

4.1.2.2 Test conditions

The tables below define the temperature and supply voltage test conditions used across the CXP00002 test stages. Each test stage references a subset of these conditions (Min, Nom, and/or Max). The CXP00002 has a single 3.3 V supply level that is shared over all the domains (Digital, PLL and Analog).

Table 5: MAG-CXP00002-NP Temperature Conditions

Condition	Symbol	Min	Nom	Max	Unit
Temperature	Temp	-40 (CT)	25 (RT)	85 (HT)	°C

Table 6: MAG-CXP00002-NP Supply Voltage Conditions

Condition	Symbol	Min	Nom	Max	Unit
Supply voltage	VDD3V3	2.97	3.3	3.63	V

In the test stage section below, the default test conditions for each stage are listed. Sometimes there is a need to include or exclude specific conditions for a particular test procedure, this will be noted in the test procedure section. If the Conditions field is empty, the default test conditions apply.

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4.1.2.3 Test stages

Production test stages:

FT (Final Test): Confirms functional and parametric compliance of the serializer. Verifies electrical parameters against specification limits after trimming. Test conditions are: CT/RT/HT and Vmin/Vnom/Vmax at the time of writing. Worst-case conditions will be set when sufficient characterization data is acquired.

Trim test stages:

TRIM @ RT, Vnom + write OTP: Trimming at room temperature and nominal voltage. Internal references are calibrated, and one-time programmable memory is written with the final trim configuration. Test conditions are: RT and Vnom.

4.1.2.4 Test procedures

The table below lists all test procedures applicable to the MAG-CXP00002-NP and indicates in which test stage each procedure is executed. The TRIM stage calibrates internal LDOs and references and programs the OTP memory, after which the FT stage verifies the full parametric and functional performance of the serializer, including its high-speed CXP interface and SPI communication channels. The Conditions column notes any deviation from the default test conditions for that test stage, an empty cell means the default conditions listed under Test stages section applies.

Table 7: MAG-CXP00002-NP Test Procedures

Test procedure	Description	Conditions	FT	TRIM
Continuity	Each pin is connected to an SMU with power supplies off. A small current is pushed into the pin to verify proper pin connectivity via the ESD diode voltage drop.		X	X
Power Short	Checks for shorts on the power rails by supplying a very low voltage and measuring the resulting current per power domain.		X	X
Pin leakage	Measures the leakage current of each input pin or tristate output pin by applying low and high voltage and measuring the current.		X	X
VIL/VIH	Checks the input hysteresis of all digital input pins by performing high-to-low and low-to-high voltage sweeps.		X	
VOH/VOL	Measures the low and high voltage of digital outputs under a load current of 1 mA.		X	
IOL/IOH	Measures the low and high level output current of digital outputs.		X	
IDDD	Measures the current consumption of the DUT in multiple states, video modes, frame rates.	FHD/HD/VGA, 10fps/25fps	X	X

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Test Buffer Offset	Measures the offset of both test buffers on the test mux by creating a loop through GPIO pins and sweeping the input voltage.		X	X
Static Signal Test	Measures all static signals available on the test mux and stores results in the database.		X	
Regulators	Performs a load test on the LDOs with external decoupling by sweeping load currents and measuring the internal output voltage through the test mux.	Vmin/Vmax	X	
Oscillator	Checks the reference clock circuits of the DUT.		X	
PLL Open Loop	Checks the PLL open loop behaviour by opening the loop and measuring the v_tune voltage range.		X	
PLL Lock	Checks the PLL closed loop behaviour by observing key parameters of the locked PLL.		X	
IO SPI Slave	Validates the debug SPI slave interface by reading and writing registers.		X	X
IO SPI Master CAM	Validates the camera SPI master interface by reading and writing a register in the FPGA acting as SPI slave.		X	
IO SPI Master GPIO	Validates the GPIO SPI master interface by reading and writing a register in the FPGA acting as generic SPI slave.		X	
CXP Uplink Receiver Sensitivity	Checks receiver sensitivity using a function generator to send CXP sequences at varying amplitudes, performing a binary search for the sensitivity threshold.		X	
Internal Test Patterns	Validates the internal test pattern generator by looping over test patterns and capturing them with the frame grabber.	FHD/HD/VGA, 8b/10b/12b, 13 test patterns	X	
Camera Test	Sends and captures test patterns from an external camera (FPGA pattern generator) and compares raw pixel values with a reference.	FHD/HD/VGA, 8b/10b/12b	X	
OTP Data Retention	Reads data from OTP and verifies that no bits are flipped.		X	X
Block Trimming	Trims each block (LDO, voltage reference, current reference) by performing a sweep and selecting the trim setting closest to the target value.			X
OTP Write	Writes the final trim configuration data into the OTP cells and verifies successful programming.			X

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The table below maps each test procedure to the datasheet parameters it verifies. Parameters are listed by their Chapter 5 name. Note that only a subset of the test procedures is used to cover all datasheet parameters.

Table 8: MAG-CXP00002-NP Parameter Traceability

Test procedure	Test ID	Parameters measured
VIL/VIH	T0004	VIL (GPIO input low threshold), VIH (GPIO input high threshold), Hysteresis (GPIO), VIL (Video input low threshold), VIH (Video input high threshold), Hysteresis (Video)
IDDD	T0007	3.3V total supply current (average), 3.3V digital supply current (average), 3.3V analog supply current (average), 3.3V PLL supply current (average), Total power consumption
Regulators	T0011	VDD1V2_CAM reference voltage
Regulators	T0012	VDD1V8_CAM supply voltage
Oscillator	T0013	Crystal frequency
Camera Test	T0023	Parallel video data rate

4.1.3 MAG-PSU00001-NP (Power Management IC)

4.1.3.1 Test flow

The production test flow for the DC/DC converter IC is depicted below. The IC undergoes connectivity (CONN) and final test (FT) after device assembly.

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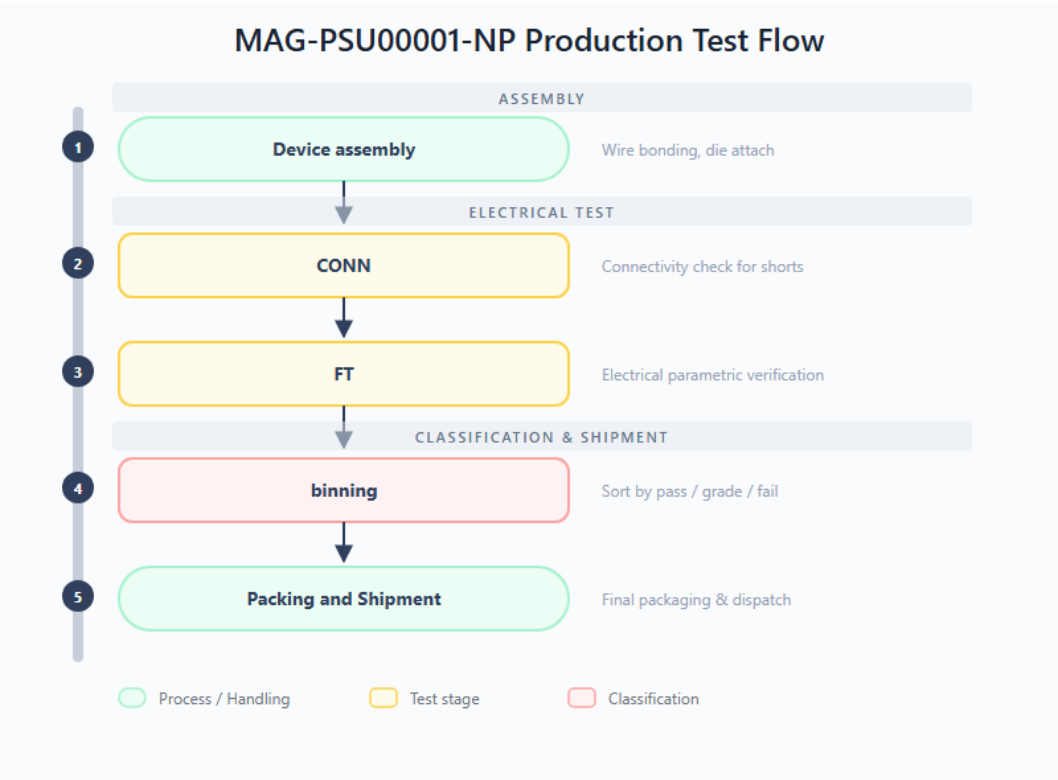

Figure 5: MAG-PSU00001-NP Production Test Flow

4.1.3.2 Test conditions

The tables below define the temperature and input voltage test conditions used across the PSU00001 test stages. Each test stage references a subset of these conditions (Min, Nom, and/or Max). The PSU00001 is a DC/DC converter with a single input voltage domain.

Table 9: MAG-PSU00001-NP Temperature Conditions

Condition	Symbol	Min	Nom	Max	Unit
Temperature	Temp	-40 (CT)	25 (RT)	85 (HT)	°C

Table 10: MAG-PSU00001-NP Input Voltage Conditions

Condition	Symbol	Min	Nom	Max	Unit
Input voltage	Vin	5	9	11	V

In the test stage section below, the default test conditions for each stage are listed. Sometimes there is a need to include or exclude specific conditions for a particular test procedure; this will be noted in the test procedure section. If the Conditions field is empty, the default test conditions apply.

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4.1.3.3 Test stages

Production test stages:

CONN (Connectivity Test): Checks for shorts on the DUT interface. DUTs must pass this stage before proceeding to ensure the test socket contact is reliable and no assembly defects are present. Test conditions are: RT and Vnom.

FT (Final Test): Verifies the chip is within specifications. Checks all electrical parameters (voltage levels, thresholds, regulation, efficiency) against defined minimum and maximum spec limits. Test conditions are: CT/RT/HT and Vmin/Vnom/Vmax at the time of writing. Worst-case conditions will be set when sufficient characterization data is acquired.

4.1.3.4 Test procedures

The table below lists all test procedures applicable to the MAG-PSU00001-NP. The PSU00001 does not include trimming or NVM programming. The CONN connectivity check is performed as a gate before the FT procedures listed here. The Conditions column notes any deviation from the default test conditions for that test stage, an empty cell means the default conditions listed under Test stages section applies.

Table 11: MAG-PSU00001-NP Test Procedures

Test procedure	Description	Conditions	CONN	FT
Power Short	Verifies that there is no short to ground on the power terminals of the chip.		X	
Load Short	Verifies that there is no short to ground or power on the load terminals of the chip.		X	
Leakage	Measures the input leakage current of all accessible input pins.			X
Input threshold	Finds both the high and low threshold voltages (VIL/VIH) for the Enable pin.			X
Output drive-strength	Measures the current that can be sunk by the power-good output.			X
On-chip regulator	Measures the regulator voltage and performs a load regulation test on the internal regulator.			X
Under Voltage Lock Out	Finds the threshold voltages where the chip enters and exits UVLO.			X
Output voltage monitor	Checks the +/-6.5% output voltage limits of the control loop by forcing the internal sense voltage.			X
Over Temperature Protection	Checks the over-temperature protection via the power-good flag and PTAT voltage. In FT test stage only the PTAT voltage is monitored.			X

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Startup behaviour	Measures the start-up behaviour of the output voltage and checks the level/time where the power-good flag goes high.	Rload: 10Ω/50Ω/270 Ω		X
Inv_Enable functionality	Checks that the functionality of the Enable pin can be inverted using the Inv_Enable configuration.			X
Line regulation	Measures the input voltage dependency of the output voltage.	Vout:0.9V/1.2 V/1.5V/1.8V/2 .0V/2.5V/3.3V , Vmin/Vmax		X
Load regulation	Measures the load current dependency of the output voltage.	Vout:0.9V/1.2 V/1.5V/1.8V/2 .0V/2.5V/3.3V , Vmin/Vmax, Iload: 0.1A/0.3A/0.5 A/0.7A/0.9A/1 .1A		X
Efficiency	Measures efficiency over different loads and input/output voltage conditions.	Vout:0.9V/1.2 V/1.5V/1.8V/2 .0V/2.5V/3.3V , Vmin/Vmax, Iload: 0.5A/0.8A/1.0 A, Halfsw:ON/OF F		X

The table below maps each test procedure to the datasheet parameters it verifies. Parameters are listed by their Chapter 5 name. Note that only a subset of the test procedures is used to cover all datasheet parameters.

Table 12: MAG-PSU00001-NP Parameter Traceability

Test procedure	Test ID	Parameters measured
Input threshold	T0006	Enable start threshold, Enable stop threshold
Under Voltage Lock Out	T0009	Vin start threshold, Vin stop threshold
Output voltage monitor	T0010	Output Over Voltage PGood threshold, Output Under Voltage PGood threshold
Startup behaviour	T0012	Soft Start duration

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4.2 Level 2: PCB Test Specifications

This section covers the functional testing of the three post-assembly sub-boards (Imager, Serializer, and Power PCBs) prior to stack integration. Each PCB is assembled with one or more pre-tested IC components from level 1. All three boards share a common test philosophy: an initial "Approved device" input gate verifies that only pre-tested level 1 components are used, followed by board-level assembly and a Vision (VIS) functional test stage that validates the sub-system operates correctly at the PCB level. The VIS test stage consists of a collection of test procedures from the component level with optional dedicated tests to increase test coverage.

4.2.1 Test flows

The production test flows for the three assembled PCBs are depicted below. While the overall structure is similar across all boards, the specific test content differs to match each sub-system's functionality.

MAG-VIS4000x-N (Imager PCB) -- Pre-tested image sensor components from level 1 are assembled onto the PCB, followed by a functional test of the board.

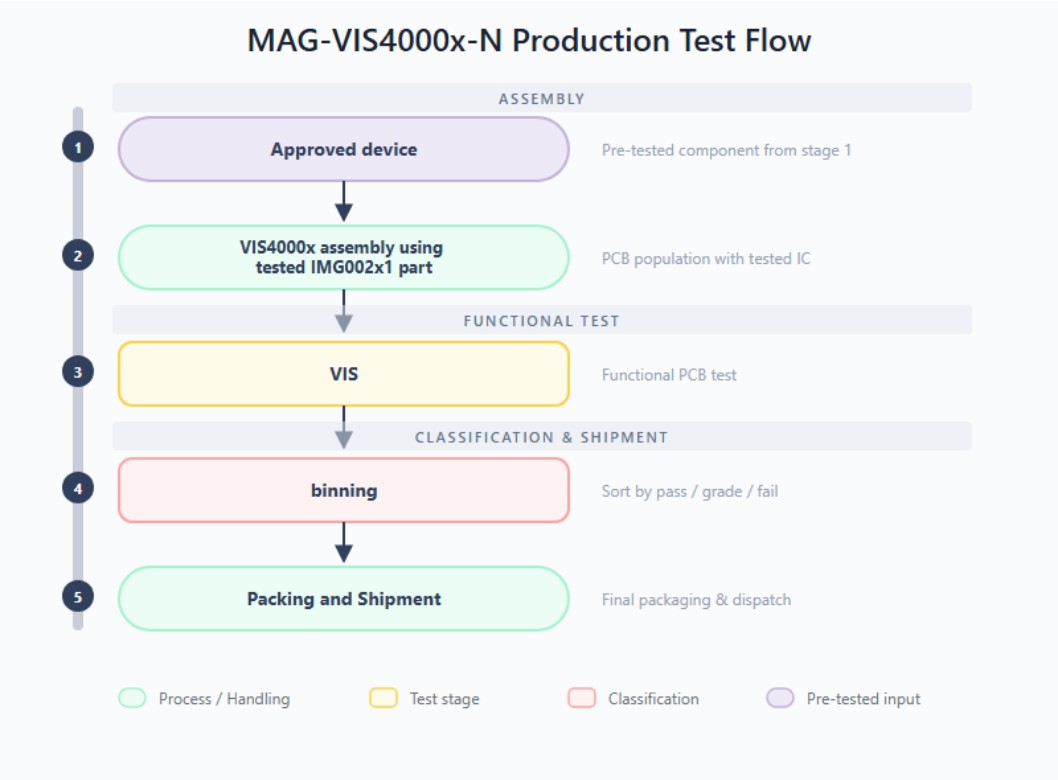

Figure 6: MAG-VIS4000x-N Production Test Flow

MAG-VIS4003x-N (Serializer PCB) -- Pre-tested serializer and DC/DC converter components from level 1 are assembled onto the PCB, followed by a functional test of the CXP link and power delivery sub-system.

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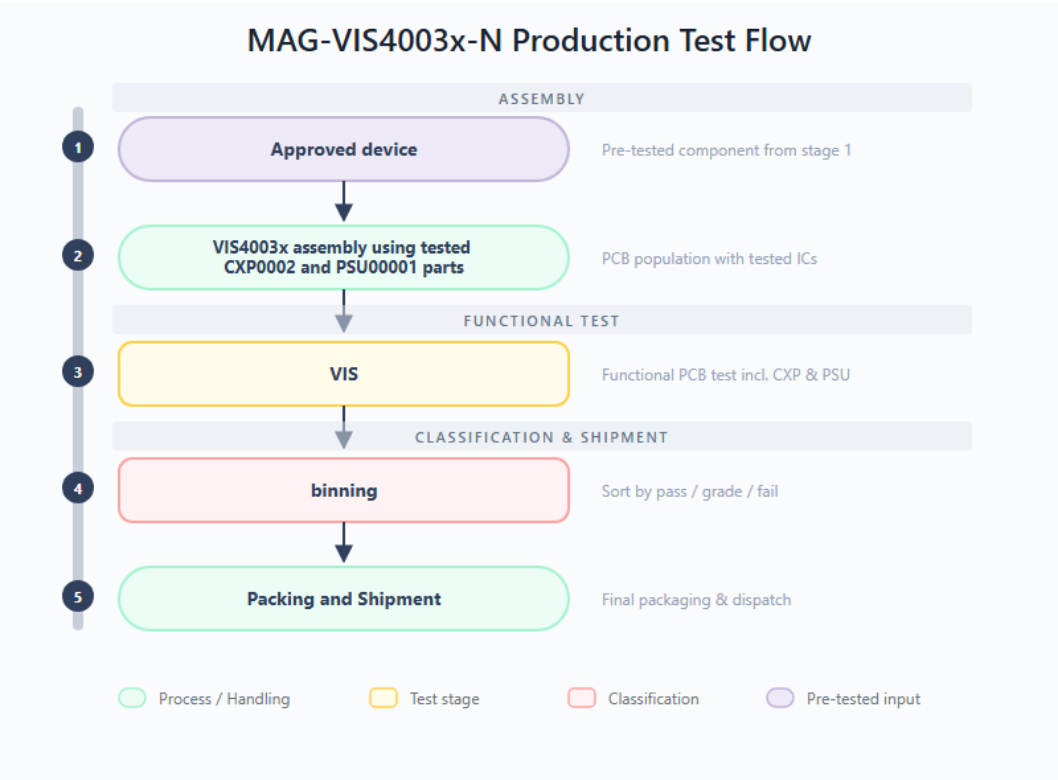

Figure 7: MAG-VIS4003x-N Production Test Flow

MAG-VIS4001x-N (Power PCB) -- Pre-tested DC/DC converter components from level 1 are assembled onto the board, followed by parametrical test to check voltage load/line regulation behaviour and ripple performance. This board additionally includes a CONN (Connectivity Test) test stage before the VIS test stage.

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Figure 8: MAG-VIS4001x-N Production Test Flow

4.2.2 Test conditions

Since the PCB-level tests are functional verification tests rather than parametric characterisation, all procedures are executed at ambient temperature and nominal supply voltage only. No temperature or supply corner conditions are applied at this level, the underlying IC components have already been fully tested at level 1.

For the MAG-VIS4000x-N (Imager PCB), the image sensor is configured in **FHD10** mode (1920 x 1080 @ 10 fps, 10-bit) during all VIS test procedures.

4.2.3 Test stages

All three PCBs share the VIS test stage as their primary functional verification. The Power PCB additionally includes a **CONN** test stage. The specific scope of each VIS test is:

> **VIS4000x -- VIS (Vision Sub-board Test):** Functional test of the assembled imager PCB. Validates that the board-level interconnects and the image sensor readout chain operate correctly after PCB population. The most important verification test in this VIS test stage is the read out of a set of test patterns that are hard coded in the image sensor IC.
> **VIS4003x -- VIS (Vision Sub-board Test):** Functional test of the assembled serializer PCB. Validates the CXP link functionality and power delivery sub-system at the board level. The most important verification test in this VIS test stage is the read-out (via CXP interface) of a set of external test patterns that are applied to the video pins.

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VIS4001x -- CONN (Connectivity Test): Checks for shorts and opens at the board level, verifying that PCB assembly and soldering have not introduced connection faults.
VIS4001x -- VIS (Vision Sub-board Test): Verifies functionality of the DCDC sub-board on the MAG-VIS400xx PCBs, including voltage-level accuracy and ripple under worst case DC conditions.

4.2.4 Test procedures

The tables below list the test procedures for each PCB during its VIS test stage. These procedures are a subset of the component-level tests, re-used at the board level to verify correct assembly and interconnect integrity. The Power PCB additionally includes a CONN stage for connectivity verification before the VIS tests.

Table 13: MAG-VIS4000x-N VIS Test Procedures

Test procedure	Description
Testpattern generator	Tests the internal testpattern generator of the image sensor IC via the PCB-level readout chain, comparing the captured image with a predefined reference.
Serial interface	Validates SPI communication with the image sensor at the board level.
IDD VIS	Measures the power supply current consumption on the VIS board, where supplies with the same voltage level are shorted and sourced as a single channel.

Table 14: MAG-VIS4003x-N VIS Test Procedures

Test procedure	Description
IO SPI Master CAM	Validates the camera SPI master interface by reading and writing a register in the FPGA acting as SPI slave.
IO SPI Master GPIO	Validates the GPIO SPI master interface by reading and writing a register in the FPGA acting as generic SPI slave.
Internal Test Patterns	Validates the internal test pattern generator by looping over test patterns and capturing them with the frame grabber.
Camera Test	Sends external test patterns (from FPGA pattern generator) to the video input pins and captures the CXP output to verify pixel-accurate data integrity.
OTP Data Retention	Reads data from OTP and verifies that no bits are flipped.
VDD3V3 DCDC	Checks the output voltage of the on-board DCDC converter.

Table 15: MAG-VIS4001x-N CONN + VIS Test Procedures

Test procedure	Description
Power Short	Verifies that there is no short to ground on the power terminals at the board level.

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Load Short	Verifies that there is no short to ground or power on the load terminals at the board level.
Line regulation	Measures the input voltage dependency of the output voltage at the board level.
Load regulation	Measures the load current dependency of the output voltage at the board level.
Efficiency	Measures efficiency over different loads and input/output voltage conditions at the board level.
Output voltage ripple	Measures ripple amplitude and frequency on the board output.

4.3 Level 3: Module Test Specifications

This section covers the Module Final Test (MFT) for the fully integrated vision module (MAG-VIS100xx-N, three-PCB vertical stack). It verifies the CoaXPress downlink/uplink and power delivery across the entire module scope, validating high-speed timing and synchronization between the different components.

4.3.1 Test flow

The production test flow for the fully integrated vision module is depicted below. Pre-tested and approved PCBs from level 2 (Imager, Power, and Serializer boards) are assembled into the final three-PCB vertical stack, followed by visual inspection and comprehensive functional verification.

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Figure 9: MAG-VIS100xx-N Production Test Flow

4.3.2 Test conditions

The tables below define the temperature, supply voltage, and imager mode test conditions used for the module-level verification. Each test procedure references a subset of these conditions.

Table 16: MAG-VIS100xx-N Temperature Conditions

Condition	Symbol	Min	Nom	Max	Unit
Temperature	Temp	0 (CT)	25 (RT)	55 (HT)	°C

Table 17: MAG-VIS100xx-N Supply Voltage Conditions

Condition	Symbol	Min	Nom	Max	Unit
Module supply voltage	VDD_IN	6	9	12	V

Table 18: MAG-VIS100xx-N Imager Mode Conditions

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Imager mode	FPS	Image size height	Image size width
FHD10	10	1080	1920
FHD30	30	1080	1920
VGA50	50	480	640

The default imager mode is FHD10 unless otherwise specified.

4.3.3 Test stages

Production test stages:

VI (Visual Inspection): Assessment of the physical condition of the assembled module to identify visible defects before electrical testing. Checks for cracks, pin issues, solder anomalies, or mechanical damage on the integrated stack.
VER (Functional Verification): Confirms that the integrated module is compliant (functional and parametric) against its specifications for normal use-cases. This stage validates CXP downlink/uplink communication, power delivery across the module, image data integrity, and trigger functionality at full operational speed. Test conditions are: RT and Vnom.

4.3.4 Test procedures

The table below lists all test procedures for the integrated module. The VI stage is a manual inspection performed before any electrical test. The VER stage validates the complete signal chain -- from power delivery through image capture to CXP serialization -- at full operational speed. Module variants that include the MAG-DRV IC have additional driver-specific tests.

Table 19: MAG-VIS100xx-N Module Test Procedures

Test procedure	Description	VI	VER
Visual inspection	Verifies the physical integrity of the assembled module prior to electrical testing, ensuring no mechanical damage, solder defects or contamination is present that could affect performance or safety.	X	
Power consumption	Measures the current consumption of the fully integrated module.		X
CXP Core functionality and video data check	Verifies basic functionality of the MAG-CXP00002-NP IC and CoaXPress video data -- checks lock bits, writes and reads a known register via CoaXPress to confirm communication, enables internal test pattern mode and verifies the correct pattern is received by the frame grabber.		X
IMG Core functionality and video data check	Verifies basic functionality of the MAG-IMG002x1 IC -- checks PLL status bits, tests image manipulation logic (y-axis flip), enables the internal Test Pattern Generator (TPG) and verifies the received image pattern via CoaXPress under various video modes and frame rates.		X

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IMG Trigger mode	Provides full functional verification of all imager trigger modes: CXP hardware trigger flow, imager free-run burst mode (30-frame burst) and imager software trigger burst mode.		X
MAG-DRV Capabilities (MAG-VIS1001x only)	Verifies functional operation of the MAG-DRV IC outputs in both steady-state (ON/OFF) and dynamic (PWM) modes, using oscilloscope and MUX card to check all channels.		X
MAG-DRV Motor drive capabilities (MAG-VIS1002x only)	Verifies the MAG-DRV IC ability to drive external loads for directional movement of Iris, Zoom and Focus motors, verifying correct drv_enable register values for both directions and safe disable (OFF state).		X

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5 Product Specifications & Acceptance Criteria

The key specifications for the Vision Series products are extracted from their respective datasheets and are listed in this chapter. The acceptance criteria and test limits are defined in the production test plans and where extracted from the test system of the product.

5.1 MAG-IMG002x1-NC (Full HD CMOS Image Sensor)

Table 20: MAG-IMG002x1-NC Key Specifications

Parameter	Datasheet Label	DS Min	DS Typ	DS Max	Unit	LSL	USL	Comment
Current consumption								
VDDD current (active, FHD25 10-bit)	I_VDDD	-	205	1000	mA	20	300	Max 1000 mA only at startup/reset
VDDA+VDDIO+VD_RST current (active, FHD25 10-bit)	I_VDDA (combined)	-	124	-	mA	20	150	Split over 3 PSU channels, LSL, USL of VDDA is given here
Clocking and control signal thresholds								
External clock reference	REFCLK	10	20	40	MHz	-	-	Fixed to 20MHz in test setup Hardware
TRIG_IN input threshold low	TRIG_in_l	0.74	1.07	1.4	V	-	-	Pin is of type: rising edge triggered
TRIG_IN input threshold high	TRIG_in_h	1.56	1.89	2.39	V	1.6	2.6	-
TRIG_IN input hysteresis	TRIG_in_hyst	0.66	0.82	1.15	V	-	-	Pin is of type: rising edge triggered
RSTB input threshold (no hysteresis)	RSTB_th_lvl	-	1.65	-	V	1.4	2.3	-
SPI interface								
SPI master clock speed	SPI_CLK	-	-	20	MHz	-	-	Fixed to 20MHz in test setup Hardware

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SPI MISO output voltage (no load)	SPI_out	2.97	3.3	3.63	V	2.7	-	Tested with 1mA load
SPI input threshold low	SPI_in_l	0.74	1.07	1.4	V	0.6	1.7	s_mosi_vil (representative)
SPI input threshold high	SPI_in_h	1.56	1.89	2.39	V	1.3	2.3	s_mosi_vih (representative)
SPI input hysteresis	SPI_in_hyst	0.66	0.82	1.15	V	0.3	0.9	s_mosi_hysteresis (representative)
Digital video output interface								
VOH (no resistive load)	VOH	2.97	VDDIO	3.63	V	2.7	-	Tested with 1mA load
VOL (no resistive load)	VOL	-	0 (gnd)	-	V	-	0.3	Tested with 1mA load
Pixel output clock speed	PIXCLK_speed	50	-	100	MHz	pass/fail	-	Only tested at speed during characterization (FHD30 mode), production testing uses FHD10, HD40 and VGA50 mode what is at half of the max speed
Data output line speed	DV_X_speed	-	-	50	MHz	pass/fail	-	Only tested at speed during characterization (FHD30 mode), production testing uses FHD10, HD40 and VGA50 mode what is at half of the max speed
Analog signals								
Bandgap reference voltage (VDD12BG)	Vvdd12bg	1.14	1.2	1.26	V	1.05	1.35	-
Electro-optical characteristics								
Effective number of pixels	-	-	1920 (H) x 1080 (V)	-	-	-	-	Guaranteed by design

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Pixel size	-	-	5um x 5um	-	-	-	-	Guaranteed by design
Number of black rows	-	-	64	-	-	-	-	Guaranteed by design
Shutter type	-	-	Rolling shutter	-	-	-	-	Guaranteed by design
Saturation limit	-	-	107000	-	e-	40000	-	Typical pre-irrad value. Related to design target full-well charge. Tested only during characterization
Conversion gain	-	-	0.012	-	mV/e-	0.006	0.018	-
System gain (K)	-	-	0.0088	-	DN/e-	0.0055	0.0165	10-bit equivalent. K = conversion gain / ADC gain
Read noise	-	-	76	-	e-	-	120	-
Dynamic range	-	-	62	-	dB	57	65	-
SNR_MAX	-	-	50.3	-	dB	46	54	-
Dark current (DC)	-	-	536	-	e-/s/px	-	2500	Typical pre-irrad value (0 Gy TID, RT)
Dark current non uniformity (DCNU)	-	-	1596	-	e-/s/px	-	4000	Typical pre-irrad value (0 Gy TID, RT)
Dark signal non uniformity (DSNU)	-	-	159	-	e-	-	600	Measured after Column FPN (CFPN) correction
Photo response non uniformity (PRNU)	-	-	1.21	-	% of mean	-2.5	2.5	Measured after Column FPN (CFPN) correction at 50% dynamic range

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Column Fixed-Pattern Noise (CFPN)	-	-	5.62	-	DN	5.175	7.075	10-bit equivalent.
Color filters	-	-	BayerRG (Bayer RGGB)	-	-	-	-	Guaranteed by design
Programmable features	-	-	Sensor parameters	-	-	-	-	Features are tested during characterization: Exposure time, sub-sampling, X-Y mirroring, ADC resolution
ADC resolution	-	-	10 and 12 bit	-	-	-	-	Guaranteed by design
Interface	-	-	10/12b parallel output	-	-	-	-	Fixed to 10b for all tests
Cover glass lid	Corning 7980 0F	-	-	-	-	-	-	Guaranteed by design

The table below lists the pixel-array defect specifications published in the MAG-IMG002x1-NC datasheet (rev 2.0, section 7.5). These limits are applicable to pre-irradiation conditions and are the datasheet acceptance criteria used during production testing in test T0035 (Optical defect map). Magics also collects a broader set of internal pixel-defect parameters (e.g. cluster counts, bright/dead row & column counts) for process monitoring; those are not listed here as they are not part of the datasheet acceptance criteria.

Table 21: MAG-IMG002x1-NC Pixel-Array Defect Specifications

Parameter	Unit	LSL	USL	Description
Number of dead pixels	px	-	50	Dead pixel is a pixel that has unusually low positive or negative response to light (stuck-at or negative-gain pixel). Dead pixels from dead rows or columns are excluded from this count.
Number of bright pixels	px	-	50	Bright pixel is a pixel with unusually high value in dark condition (high outliers in the pixel DC histogram, or broken pixel/col/row with unusually high offset). Bright pixels from bright rows or columns are excluded from this count.
Number of bad columns	cols	-	0	Bad_col = dead_col + bright_col. USL = 0 for production (PROD) parts; parts with 1–5 bad columns are re-binned as engineering model (EM); >5 bad columns → part is rejected.

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Number of bad rows	rows	-	0	Bad_row = dead_row + bright_row. USL = 0 for production (PROD) parts; parts with 1–5 bad rows are re-binned as engineering model (EM); >5 bad rows → part is rejected.
Distance between bad columns	cols	5	-	Only applicable for EM parts.
Distance between bad rows	rows	5	-	Only applicable for EM parts.

5.2 MAG-CXP00002-NP (CoaXPress Serializer ASIC)

Table 22: MAG-CXP00002-NP Key Specifications

Parameter	Datasheet Label	DS Min	DS Typ	DS Max	Unit	LSL	USL	Comment
Current consumption								
3.3V total supply current (average)	IVDD3V3_TOTAL_AVG	-	82.35	-	mA	-	-	Sum of analog + digital
3.3V digital supply current (average)	IVDD3V3_DIG_AVG	14.45	18.6	28	mA	20	100	Channel combines DIG+PLL
3.3V analog supply current (average)	IVDD3V3_ANA_AVG	-	49.8	-	mA	40	60	Active mode measurement
3.3V PLL supply current (average)	IVDD3V3_PLL_AVG	-	13.93	-	mA	-	-	Included in DIG, no separate Power supply channel
Power consumption								
Total power consumption	-	-	271.55	-	mW	-	-	Derived from IDD x VDD
Logic level inputs (GPIO)								

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VIL (GPIO input low threshold)	VIL	390	470	635	mV	390	635	-
VIH (GPIO input high threshold)	VIH	443	550	715	mV	443	715	-
Hysteresis (GPIO)	Hysteresis	38	80	117	mV	38	117	-
Logic level inputs (Video pins)								
VIL (Video input low threshold)	VIL	1.147	1.328	1.689	V	1.147	1.689	-
VIH (Video input high threshold)	VIH	1.482	1.714	2.282	V	1.482	2.282	-
Hysteresis (Video)	Hysteresis	309	387	593	mV	38	593	-
Data rates								
Parallel video data rate	DRin,par	-	-	100	MHz	pass/fail	-	Functional test
Crystal oscillator								
Crystal frequency	fXTAL	10	20	40	MHz	19.5	20.5	-
CAM LDO outputs								
VDD1V8_CAM supply voltage	V_VDD1V8_CAM	1.7	1.8	1.9	V	1.2	2.4	-
VDD1V2_CAM reference voltage	V_VDD1V2_CAM	1.15	1.2	1.25	V	1.15	1.25	-

5.3 MAG-PSU00001-NP (10W Synchronous Step-Down DC/DC Converter)

Table 23: MAG-PSU00001-NP Key Specifications

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Parameter	Datasheet Label	DS Min	DS Typ	DS Max	Unit	LSL	USL	Comment
Power								
Input voltage supply range	PVin, Vin	5	-	11	V	-	-	Test condition
Output current (PCB in air)	Iout	-	-	1	A	-	-	Test condition
Output power (PCB in air)	Pout	-	-	2	W	-	-	Test condition
PWM								
Maximum Duty Cycle	DMax	-	100	-	%	-	-	Cannot be measured, sensitive node
Minimum Duty Cycle	DMin	-	0	-	%	-	-	Cannot be measured, sensitive node
Error Amplifier								
DC Gain	DCG	-	90	-	dB	-	-	Cannot be measured, sensitive node
Unity Gain-Bandwidth	UGBW	-	20	-	MHz	-	-	Cannot be measured, sensitive node
Slew Rate	SR	-	10	-	V/us	-	-	Cannot be measured, sensitive node
Under-voltage lockout								
Vin start threshold	VinStartTh	-	4.79	-	V	4.4	4.8	-
Vin stop threshold	VinStopTh	-	4.52	-	V	4.12	4.53	-
Enable								

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Enable start threshold	EnStartTh	-	815	-	mV	770	900	-
Enable stop threshold	EnStopTh	-	730	-	mV	670	780	-
Enable pin series resistance	EnSerRes	-	10	-	kΩ	-	-	Guaranteed by design
Protections								
Over Current Protection peak level	OCPpk	-	6	-	A	-	-	Cannot be triggered with test setup, functionality of OCP is not covered within test plan
Over Current Protection average level	OCPavg	-	4.8	-	A	-	-	Cannot be triggered with test setup, functionality of OCP is not covered within test plan
Over Temperature start threshold	OTPStartTh	-	103	-	°C	-	-	Only tested during characterization
Over Temperature stop threshold	OTPStopTh	-	73	-	°C	-	-	Only tested during characterization
Soft Start								
Soft Start duration	SSt	-	-	440	us	-	1500	-
Power Good								
Output Over Voltage PGood threshold	OV	-	+6.5	-	%	2.5	10.5	-
Output Under	UV	-	-6.5	-	%	-10.5	-2.5	-

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Voltage PGood threshold								
PTAT								
PTAT temperature slope	PTAT	-	8.5	-	mV/°C	-	-	Only tested during characterization

5.4 MAG-VIS100xx-N (Electronic Vision Module)

Table 24: MAG-VIS100xx-N Key Specifications

Parameter	Datasheet Label	DS Min	DS Typ	DS Max	Unit	Comment
Electrical characteristics						
Module supply current	I_SUPPLY_9V	160	-	257	mA	-
AC output impedance (HD-BNC)	Z_out_AC	-	75	-	Ohm	-

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6 References

6.1 Datasheets

Table 25: Datasheet Reference Documents

Document Name	Version
MAG-IMG002x1-NC Datasheet	Rev. 2.0 (April 2026)
MAG-CXP00002-NP Datasheet	Rev. 2.2 (April 2026)
MAG-PSU00001-NP Datasheet	Rev. 1.0 (April 2026)
MAG-VIS100xx-N Datasheet	Rev. 1.2 (April 2026)

6.2 Product test plans

The detailed per-product production test plans listed below are internal Magics documents used as input for the present overview document. They define the device-under-test fixtures, equipment, pin-mapping, and the step-by-step test procedures that implement the test stages described in Chapter 3.

Table 26: Product Test Plan Documents

Product	Test Plan Document	Revision
MAG-IMG002x1-NC	IMG002x1_test_plan_1v2.docx	Rev. 1.2
MAG-CXP00002-NP	mag-cxp-product-test-plan_1v2.docx	Rev. 1.2
MAG-PSU00001-NP	ProductTestDevelopmentMAG-PSU_test_plan_1v2.docx	Rev. 1.2
MAG-VIS100xx-N	MAG_VIS100XX_EVK_Modules_Test_Plan.docx	TBD

6.3 Product specification and test Limits-file GIT traceability

Each product has a dedicated GIT repository that contains the full software of the production test system used to test that product, including the test sequences, post-processing scripts and the Product specification and test Limits-file. The Limits-file holds the acceptance limits (LSL/USL) applied by the test system to decide pass/fail and binning of every device under test. It is the single source of truth used at run-time; any change to a limit is traceable through the commit history of this file. The repositories are hosted under the `magics-tech` GitHub organization (`https://github.com/magics-tech/<repo>`).

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The table below pins, per product, the repository hosting the test-system software, the active branch, the path to the Product specification and test Limits-file inside the repository, and the last commit that updated that file.

Table 27: Product specification and test Limits-file GIT Traceability

Product	GIT repository (test-system software)	Branch	Limits-file path	Last commit on Limits-file	Commit date	Commit message
MAG-IMG002x1-NC	magics-tech/Vision-MEGACAM-python-sw	Product_tester	vision_megacam_python_sw/f_post_processing/check_limits.xlsx	38774b6	2026-04-24	Update Limits-file with new limit configurations for FPN correction
MAG-CXP00002-NP	magics-tech/proxima-product-test-sw	master	sw/f_post_processing/check_limits.xlsx	28f7e3e	2026-01-21	Update limits of M0103, M0133, M0135 and M0137 based on testing results of 22 parts. Set vref_ptat to force pass since this circuit shows unstable behavior but this can be waived since not used in application
MAG-PSU00001-NP	magics-tech/DCDC-tester-python-sw	master	dcdc_tester_python_sw/f_post_processing/check_limits.xlsx	c76063b	2026-03-23	added condition to limit for V_PTAT

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MAG-VIS100xx-N	magics-tech/vision-evk-product-tester-python-sw	master	Not yet present in repo	-	-	Limits-file to be added once defined

7 Glossary / Abbreviations

Abbreviation	Definition
ADC	Analog-to-Digital Converter
ATE	Automatic Test Equipment
CONN	Connectivity Test
CT	Cold Temperature
CXP	CoaXPress (high-speed serial interface standard)
DC	Dark Current
DCF	Device Configuration File
DCNU	Dark Current Non-Uniformity
DSNU	Dark Signal Non-Uniformity
ESD	Electrostatic Discharge
FHD	Full High Definition (1920 x 1080)
FPN	Fixed Pattern Noise
FPS	Frames Per Second
FT	Final Test
GPIO	General Purpose Input/Output
HD	High Definition (1280 x 720)
HT	High Temperature
LDO	Low Dropout Regulator
LSL	Lower Specification Limit
MFT	Module Final Test
NVM	Non-Volatile Memory
OT	Optical Test
OTP	One-Time Programmable (memory)
PCB	Printed Circuit Board

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PLL	Phase-Locked Loop
PRNU	Photo Response Non-Uniformity
PTAT	Proportional To Absolute Temperature
PTC	Photon Transfer Curve
ROI	Region Of Interest
RT	Room Temperature
SMU	Source Measure Unit
SN	Spatial Non-uniformity
SPC	Statistical Process Control
SPI	Serial Peripheral Interface
TPG	Test Pattern Generator
TRIM	Trimming (calibration of internal references)
USL	Upper Specification Limit
UVLO	Under-Voltage Lock-Out
VCO	Voltage-Controlled Oscillator
VER	Functional Verification (module test stage)
VGA	Video Graphics Array (640 x 480)
VI	Visual Inspection
VIS	Vision Sub-board Test

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